

The humus balance model (HU-MOD): a simple tool for the assessment of management change impact on soil organic matter levels in arable soils

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Abstract With this paper we present a simple model for the assessment of management impact in arable farming systems on soil organic matter (SOM) levels. The humus balance model (HU-MOD) is designed for application by farmers and extension workers in practice as a tool for management support. To enable practice applicability, HU-MOD bypasses the need for data on soil parameters and can be run with simple management data. HU-MOD is based on a simplified model on carbon and nitrogen pools and fluxes in the soil–plant system. The model proved to be an applicable simple tool for the comparison of management systems in arable farming with regard to the impact on SOM levels. Even though an absolute quantification of SOM level changes is not possible due to the methodical approach bypassing the need for any data on soil parameters, the model may be used to assess a positive or negative impact of a management system or management period compared to a reference and thus may be used to assess the impact of management changes, or to analyse a specific impact for different management periods on a defined spatial unit.

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List of symbols

DOK	DOK long-term field experiment
HBA	Humus balance
HRC	Humus reproduction coefficient
HU-MOD	The HUmus balance MODel
LTFE	Long-term field experiment
OAFEG	Organic arable farming experiment gladbacherhof
SEE	Standard error of the estimate
SOC	Soil organic carbon
SOM	Soil organic matter
SON	Soil organic nitrogen
STN	Soil total nitrogen

Introduction

An appropriate management of soil organic matter (SOM) is a major challenge for sustainable agriculture. Being a strong determinant of numerous processes providing soil services, SOM is a key factor of soil quality (Lal 2006) and has environmental as well as agronomic relevance. Carbon sequestration possibly is the most recognised environmental service which is provided by SOM (Freibauer et al. 2004). With respect to agriculture, beneficial SOM services further include storage, transformation and supply of nutrients, amelioration of soil structure, and filter and

buffer processes (Reeves 1997; Lal 2006). Adequate SOM management is therefore a measure of eco-functional intensification of agricultural systems and can, besides other benefits e.g. for soil structure, substitute the application of synthetic inputs, especially nitrogen fertilizers (Nissen and Wander 2003). This conclusion is relevant for organic farming in particular, but should also be strongly considered with the improvement of integrated farming strategies.

Regarding the assessment of SOM management in farming practice on the field, farm, or regional scale, there is a need for tools that get along with commonly available input data, as are crop rotation, yields, and type and amount of applied fertilizer. Spatially representative data on SOM parameters are difficult to collect and will usually not be available. Furthermore, state assessment of SOM level indicators is lacking applicable reference values up to now (Wessolek et al. 2008; Loveland and Webb 2003). For this reason, modelling approaches must be considered preferable. However, models may not require too much expert knowledge, if they shall be applied by farmers and extension workers. With this paper, we introduce the Humus Balance Model (HU-MOD)¹ that can be run completely independent of soil parameter measurements and requires only easily available management data as an input. The model is based on the “horizontal nitrogen balance” (Leithold 1991) and thus has a conceptional link to the humus balance approach of Körschens et al. (2004) that is even recognized with the legislation on agricultural subsidies in Germany (DirektZahlVerpflV 2004). HU-MOD is a fully applicable model, but the tool may also be used to calculate coefficients for the method of Körschens et al. (2004).

In contrast to other models on SOM dynamics, HU-MOD does not aim at the absolute quantification of SOM dynamics, as this would not be possible independent of soil data. Instead, HU-MOD calculates specific humus reproduction levels² for arable cropping

systems that can be related to each other. It is hypothesized that the variation between different humus reproduction levels at one site is positively correlated to the variation SOM level changes, independent of the magnitude and general trend of the latter. Therefore, the model may be used to assess the impact of management changes, or, retrospectively, to compare humus reproduction levels of different past management periods to each other. Up to now, the model may not be used to quantify absolute SOM level changes, but may be used for the assessment of a dimensionless positive or negative impact on SOM levels of a management system compared to the specific reference system.

Materials and methods

Model description

The model aims at the calculation of coefficients that relate to the specific positive or negative impact of arable crops on SOM stocks as a combined result of nutrient uptake, cultivation impact, and organic matter input in a cropping system. The generation of humus reproduction coefficients (HRC) in HU-MOD is based on a simplified model of pools and fluxes of nitrogen and carbon in the soil–plant system (Fig. 1). The model concept in principal relates to the “horizontal nitrogen balance” (Leithold 1991), and to the further development of Leithold’s approach by Hülsbergen (2003).

Roughly, humus reproduction coefficients are generated for every crop in the survey period based on the calculation of management induced SOM loss and SOM supply.

$$\text{HRC} = \text{SOM}_{\text{SUP}} - \text{SOM}_{\text{LOSS}} \quad (1)$$

HRC = humus reproduction coefficient (kg SOC).

SOM_{SUP} = supply of organic material effectively contributing to SOM build-up (kg SOC).

SOM_{LOSS} = management induced SOM loss as a consequence of the absorption of mineralized SON

¹ The authors lately became aware that Malkina-Pykh and Pykh (1998) presented a model on soil organic matter dynamics named HUMOD. However, as our model has been introduced to the scientific community and the extension sector in Germany as HUMOD, we decided to insert the hyphen instead of changing the name.

² The term of “humus reproduction” is widely used with the concept of humus balances (e.g. Asmus and Herrmann 1977; Leithold 1991; Körschens et al. 1998; Hülsbergen 2003; Kolbe 2010; Engels et al. 2010; Franko et al. 2011) and relates the

Footnote 2 continued

supply of management induced SOM loss in arable cropping systems by organic matter inputs.

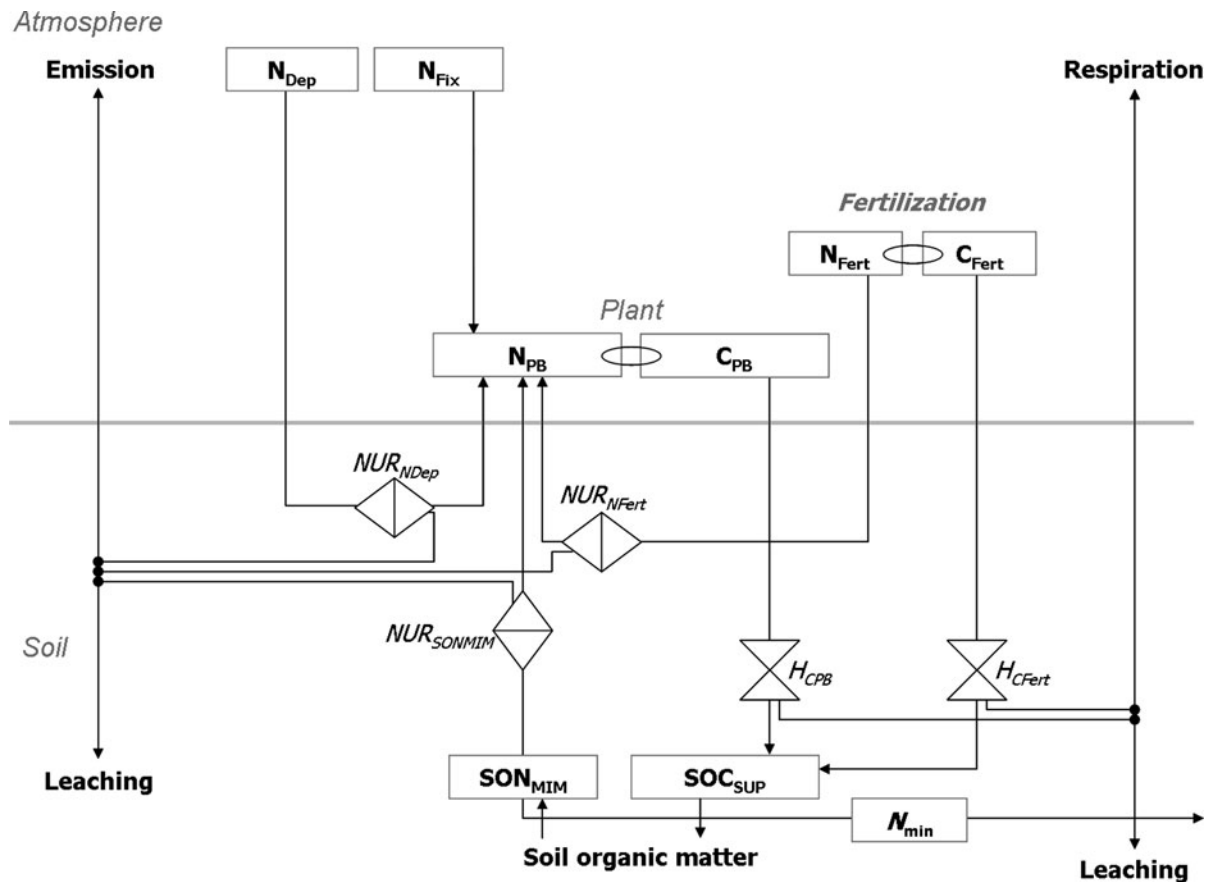


Fig. 1 Conceptual scheme of the humus balance model HU-MOD. Boxes refer to pools that are parametrized in the model. Pools without box are not parametrized. Associated pools are denoted by oval connectors. Valves refer to organic carbon turnover, parallelograms refer to organic and inorganic nitrogen turnover. SOC soil organic carbon, SON soil total nitrogen. SON_{MIM} Management induced SON loss. SOC_{SUP}

supply of organic material effectively contributing to SOM build-up. N_{PB} , C_{PB} nitrogen (N) and carbon (C) in plant biomass, N_{Fert} , C_{Fert} nitrogen (N) and carbon (C) in fertilizers, N_{Dep} nitrogen input by atmospheric deposition. N_{Fix} nitrogen input by symbiotic fixation of legumes. N_{min} (residual) nitrogen in soil solution. NUR nitrogen utilization rate (pool specific). H humification rate (input specific)

by crops, and the promotion of SOM mineralization by mechanical impact (kg SOC).

Management induced SOM loss is estimated relating to nitrogen in plant biomass (Fig. 1).

According to that scheme, humus demand equals the minimal amount of SOM that has necessarily been mineralized to supply the amount of nitrogen incorporated into a defined plant biomass, considering the possible contribution of other input pools to nitrogen supply of plants as well as unproductive SOM mineralization as a result of intensive tillage measures. Partial utilization of nitrogen supply from different pools by plants is recognized by applying input-

specific utilization rates for N that are modified according to site conditions.

In mathematical terms, humus demand is calculated as:

$$SOM_{LOSS} = SON_{MIM} \times CF \quad (2)$$

SOM_{LOSS} = management induced SOM loss as a consequence of the absorption of mineralized SON by crops, and the promotion of SOM mineralization by mechanical impact (kg SOC).

SON_{MIM} = management-induced SON mineralization (kg SON).

CF = conversion factor SON to SOC (factor).

$$\text{SOM}_{\text{MIM}} = \frac{N_{\text{PB}} - N_{\text{Fix}} - N_{\text{Dep}} \times \text{NUR}_{\text{NDep}} - N_{\text{Fert}} \times \text{NUR}_{\text{NFert}}}{\text{NUR}_{\text{SONMIM}}} + \Delta N_{\text{min}} \quad (3)$$

SON_{MIM} = management-induced SON mineralization (kg SON).

N_{PB} = nitrogen in plant biomass (kg N).

N_{Fix} = nitrogen derived from the atmosphere by legumes via symbiotic fixation (kg N).

N_{Dep} = nitrogen from atmospheric deposition (kg N).

N_{Fert} = nitrogen from fertilizers (kg N).

NUR = nitrogen utilization rates (%), pool-specific. Modified according to site quality and yield level.

ΔN_{min} = excessive nitrogen mineralization due to mechanical impact (kg N).

Humus supply by a crop is estimated relating to carbon input with organic matter in a cropping system (Fig. 1), taking into account root components, exudates, and harvest residues, as well as organic matter from fertilizers (straw or green biomass inputs, animal fertilizers, and other organic fertilizers).

Effective SOM build-up by organic inputs is calculated by applying substrate-specific humification rates.

$$\text{SOC}_{\text{SUP}} = C_{\text{R}} \times H_{\text{R}} + C_{\text{RT}} \times H_{\text{RT}} + C_{\text{EX}} \times H_{\text{EX}} + C_{\text{RE}} \times H_{\text{RE}} + C_{\text{OF}} \times H_{\text{OF}} \quad (4)$$

SOC_{SUP} = effective SOC supply with plant biomass (kg SOC).

$C_{\text{R}}, C_{\text{RT}}, C_{\text{EX}}, C_{\text{RE}}, C_{\text{OF}}$ = C-input (kg C) with roots at harvest (R), roots decayed during plant growth (RT), root exudates (EX), aboveground residues (RE), and organic fertilizers of any kind (OF).

H = humification coefficients (factors), substrate-specific with R, RT, EX, RE, and OF, respectively.

To allow for integration of the C-based and N-based algorithms, the values are transformed to kg SOC according to Rauhe and Schönmeier (1966).

The humus balance (HBA) is calculated constitutively as the sum or the mean of HRC in the survey period, depending on the intended assessment level.

Model setup and parameterization

HU-MOD is supposed to run with very little input data requirements in order to achieve applicability as a practical decision support tool. This means that parameterization must be possible based on commonly available data in farming practice, as are crop type, crop yield, and type and amount of applied fertilizers. A very basic calibration is conducted by considering the crop yield level and site quality level (see below). More detailed calibration is possible, but not compulsory.

SOM loss sub-algorithm

Nitrogen content in plant biomass (N_{PB}) is calculated according to:

$$N_{\text{PB}} = \text{MP}_{\text{FM}} \times \text{MP}_{\text{DM}} \times \text{MP}_{\text{N}} + \text{SP}_{\text{FM}} \times \text{SP}_{\text{DM}} \times \text{SP}_{\text{N}} + \text{RE}_{\text{DM}} \times \text{RE}_{\text{N}} + \text{R}_{\text{DM}} \times \text{R}_{\text{N}} \quad (5)$$

N_{PB} = nitrogen in plant biomass (kg N)

MP_{FM} = main product fresh matter yield (kg FM)

MP_{DM} = main product dry matter content (ratio)

MP_{N} = main product dry matter N content (ratio)

SP_{FM} = side product fresh matter yield (kg FM)

SP_{DM} = side product dry matter content (ratio)

SP_{N} = side product dry matter N content (ratio)

RE_{DM} = unharvestable aboveground residue dry matter (kg DM)

RE_{N} = unharvestable aboveground residue dry matter N content (ratio)

R_{DM} = root dry matter (kg DM)

R_{N} = root dry matter N content (ratio)

To achieve applicability, all parameters except for main product fresh matter yield (MP_{FM}) are calculated based on default values obtained from comprehensive German agricultural data collections (Hülsbergen 2003; KTBL 2005). Values for some selected crops are given in Table 1. Side product fresh matter yields can be calibrated if data are available. If this is not the

Table 1 Initial parametrization of the HU-MOD algorithm for selected crops

Crop	SP	R	RE	DMMP	DMSP	NMP	NSP, NRE	NR	MPC	SPC, REC	N _{FIX}	ΔN _{min}
	kg kg ⁻¹ MP	kg kg ⁻¹ SB	kg kg ⁻¹ R	kg kg ⁻¹ FM	kg kg ⁻¹ FM	g kg ⁻¹	g kg ⁻¹	g kg ⁻¹	g kg ⁻¹	g kg ⁻¹	g N kg ⁻¹ N _{PB}	kg N ha ⁻¹
Clover-grass	NA	2.00	2.50	0.20	NA	26.00	26.00	18.60	460	460	630	0.00
Winter wheat	1.00	2.00	2.50	0.86	0.86	18.00	4.50	8.00	450	460	0.00	0.00
Winter rye	1.20	1.36	2.80	0.86	0.86	18.00	4.50	8.00	450	460	0.00	0.00
Summer barley	1.0	1.54	2.20	0.86	0.86	18.00	4.50	8.00	450	460	0.00	0.00
Oats	1.4	1.54	2.60	0.86	0.86	17.00	4.00	8.00	460	465	0.00	0.00
Sugar beet	0.6	9.90	NA	0.23	0.18	7.00	18.00	26.30	410	400	0.00	25.00
Maize (silage)	NA	8.89	1.30	0.28	NA	13.00	13.00	8.00	450	450	0.00	0.00
Potatoes	0.30	5.60	NA	0.22	0.40	14.00	9.00	23.00	420	400	0.00	50.00
Peas	1.50	9.80	2.00	0.86	0.86	37.00	16.00	25.00	470	475	600	0.00
Soy	1.60	6.10	1.50	0.86	0.86	47.00	13.80	11.80	470	475	700	0.00
Catch crop	NA	2.00	2.50	0.18	NA	30.00	30.00	18.60	460	460	630	−30.00

MP main product, SP side product, R roots at harvest, SB shoot biomass (DMMP + DMSP), RE aboveground residues, DMMP main product dry matter, DMSP side product dry matter, NMP main product dry matter N content, NSP, NRE side product/aboveground residue dry matter N content, NR root dry matter N content, MPC main product dry matter C content, SPC/REC side product/aboveground residue dry matter C content, N_{FIX} N derived from the atmosphere by legumes via symbiotic fixation, ΔN_{min} exceptional change in soil mineral N under cropping systems

case, SP_{FM} should be calculated based on a default MP_{FM}:SP_{FM} ratio. The calculation of SP, RE and R is described below with the SOM supply algorithm.

If available, the model can be calibrated using regionally adapted data collections. A variation of N contents in plants as a result of different nutrient supply is only considered by a static differentiation between conventional and organic farming, the latter usually achieving lower N contents of non-legume crops (KTBL 2005; Casagrande et al. 2009).

Symbiotic nitrogen fixation (N_{FIX}) was quantified according to Hülsbergen (2003) (Table 1). In legume-/non-legume-mixtures, N_{FIX} is only calculated for legume crops. N-transfer from legume to non-legume crops (Brophy et al. 1987, Fustec et al. 2010) has not been included up to the present, partly because of the vast difficulties in assessing and quantifying the N transfer (Høgh-Jensen 2006). A negative correlation between symbiotic nitrogen fixation of legumes and readily available nitrogen in soil solution as reported by Möller et al. (2008) is considered by a decrease of N_{FIX} that is assumed to be proportional to N_{Fert} increase. Of course, N_{FIX} may not reveal negative values.

An intensive mechanical impact of a cropping system on soils (e.g. potato cropping, cf. Angers et al. 1999) in the model is supposed to promote excessive mineralization of SON with regard to plant nitrogen uptake. Such cases are displayed by positive ΔN_{min} values, thus increasing the calculated SOM loss of the respective cropping system. On the other hand, crops with a special ability to preserve residual N_{min} from leaching obtain negative ΔN_{min} values to express a lower demand of nitrogen from SON mineralization.

For the time being, ΔN_{min} was only specified with preliminary portions for some crops (Table 1).

Deposition of nitrogen from the atmosphere (N_{Dep}) shows a large spatial variability (Stein-Bachinger et al. 2004). Following Hülsbergen (2003) we apply 20 kg N ha⁻¹ a⁻¹ as standard amount, which is supported by findings from Weigel et al. (2000) or Lipavský et al. (2008) for mid-european landscapes.

The model parameter N_{Fert} refers to the quantity of fertilizer nitrogen that becomes available for plants in the respective cropping period. Calculation of N_{Fert} is based on data on the proportion of mineral N and short-term available N in different organic fertilizers

Table 2 Parametrization of nitrogen supply to plants by organic fertilizers

Fertilizer	N_{Fert} (kg N kg ⁻¹ N _{TI})	N_{Fym} (kg N kg ⁻¹ N _{TI})
Dung water (cattle)	0.81	NA
Slurry (cattle)	0.38	NA
Slurry (pigs)	0.60	NA
Fresh solid stable manure (cattle)	NA	0.21
Rotten solid stable manure (cattle)	NA	0.17
Compost	NA	0.21

N_{Fert} N supply of fertilizers with a high share of short-term available N, N_{Fym} N supply of fertilizers with a high share of stabilized N. N_{TI} total N input with fertilizer, NA not applicable

as reported by Stein-Bachinger et al. (2004) as well as Hülsgen (2003) (Table 2). Application losses are considered according to Hülsgen (2003), assumed to account for approx. 15% of short-term available N, without further differentiation.

The partial utilization of nitrogen from different source pools by plants is expressed by the nitrogen utilization rates (NUR):

$$NUR = \frac{N_{uptake}}{N_{input}} \quad (6)$$

NUR = nitrogen utilization rate (%).

N_{uptake} = specific uptake of nitrogen from an input pool by a plant (kg N).

N_{input} = nitrogen input from a defined source (kg N).

Simplified input-specific utilization rates for N are defined based on results of Leithold (1982), Rauhe et al. (1987), and Hülsgen (2003), considering

different release dynamics of N from different pools. In HU-MOD, NUR for N_{Dep} and N_{Fert} are modified depending on site conditions and yield level (Table 3). However, the modification applies only to N from fertilizers with high proportions of readily available N (mineral fertilizer, slurry), while NUR for fertilizer-N from a slow continuous source (solid manure, organic stabilization of N) is fixed preliminarily at 0.9. As the latter NUR only refers to the organic manure nitrogen that becomes available for the plants in the year of application (*cf.* Table 2), calculated N supply to the crops from organic fertilizers applying this value is still in line with the range of mineral fertilizer equivalents (MFI) for solid manure reported by Gutser et al. (2005) in their review. The utilization of nitrogen from soil organic matter mineralization by plants (NUR_{NH}) is assumed to be on a comparable level as with farmyard manure.

An impact of site conditions on nitrogen utilization by plants is recognized by applying four site quality levels (Table 4). The underlying assumption is that a higher site quality promotes a better utilization of available nitrogen resources by plants. For application in Germany, the “Ackerzahl” is used as site quality index. Ackerzahl ranges correlated to the four site quality levels are shown in the table.

Connecting NUR to yield levels is based on the assumption that the yield level at a given site is positively correlated to nitrogen utilization by plants. With regard to NUR alteration, yields in HU-MOD are classified into crop-specific “low”, “average” and “high” levels.

SOM supply sub-algorithm

Quantification of the supply pools was conducted based on the outcomes of a comprehensive review of

Table 3 Site quality levels to be applied for the estimation of N utilization rates (NUR)

Site quality level	Corresponding Ackerzahl index	Description
Poor	<25	Sites with an inherent low yield potential, as are sandy soils in regions with low precipitation.
Fair	25...50	Sites with a rather low yield potential, but higher than “poor” sites. E.g. Loamy soils with suboptimal aeration and/or mean annual temperature and/or precipitation.
Good	50...75	Sites with a sufficient yield potential, e.g. loamy soils with sufficient aeration, precipitation and mean annual temperature.
Very good	>75	Sites with a high inherent yield potential, as loess soils with optimal aeration, water supply and mean annual temperature.

Table 4 Default nitrogen utilization rates (NUR) depending on site quality and yield level

Yield level	Site quality level			
	Poor	Fair	Good	Very good
Low	0.40	0.50	0.60	0.70
Medium	0.45	0.55	0.65	0.75
High	0.50	0.60	0.70	0.85

Rates are applied for N from sources with high amounts of readily available mineral N (atmospheric deposition, mineral nitrogen fertilizers, slurry, dung water, etc.)

Table 5 Default ratios for the calculation of residue input of crops

Crop	Ratios			
	SR quotient	RRE quotient	RRT quotient	REX quotient
Clovergrass	2.00	2.50		
Winter wheat	2.00	2.50		
Potatoes	5.60	0.00	2.00	0.65
Peas	9.80	2.00		
Winter rye	1.36	2.80		

Ratios: *SR* shoot-to-roots at harvest, *RRE* roots at harvest-to-aboveground residues, *RRT* roots at harvest-to-roots decayed during vegetation period, *REX* roots at harvest-to-root exudates

the literature, aiming at the identification of appropriate ratios to derive C-pool magnitudes based on crop yields. Default values applied in the SOC_{SUP} algorithm are shown for some example crops in Table 5.

Crop specific shoot:root ratios for calculation of CR where defined based on Körschens et al. (1989), Klimanek (1990, 1997), Bolinder et al. (1997, 2007), Jost (2003), KTBL (2005), and Hülsbergen (2003). It has to be noted that “shoot” in the calculation refers to crop yield, excluding aboveground harvest residues.

Aboveground harvest residues (C_{RE}) were estimated by applying another crop specific ratio. This ratio was derived relating to Körschens et al. (1989), and Hülsbergen (2003).

Root turnover (C_{RT}), a parameter recognizing the biomass input of roots died off during the vegetation period, was determined to account for double the amount of root biomass at harvest according to Swinnen et al. (1995), Steingrobe et al. (2001), and Rasse et al. (2005).

The organic C input with root exudates (C_{EX}) was calculated based on the assumption that about 1/3 of relocated plant C is released with root exudates, whereas about 1/2 will remain in roots (Kuz'yakov and Domanski 2000; Kuzyakov and Schneckenberger 2004).

Humification rates in HU-MOD (H) are applied as substrate specific static values.

Preliminarily, they are postulated not to change according to site conditions (site quality level) or any other factor. The contribution of root-C to SOM build-up is assumed to be 1.5–3 times higher than the one of aboveground plant material (Allmaras et al. 2004; Johnson et al. 2006). Based on results of Hansen et al. (2004) and Parshotam et al. (2000), the humification coefficient for root material (H_R) is set to 0.25 in HU-MOD, without further differentiation.

The root turnover input pool in HU-MOD (C_{RT}) is comprised mainly of fine roots, root hairs and dissociated plant cells. Since all of these materials are rather easily decomposable (Swinnen et al. 1995; Steingrobe et al. 2001), the humification coefficient H_{RT} is defined as 0.14.

Root exudates are only provided with a humification coefficient H_{EX} of 0.05 in HU-MOD. This is due to the fact that a large proportion 62–86% of exudates are dissimilated quickly (Merbach and Wittenmayer 2004; Kaštovská and Šantrůčková 2007), whereas only a small proportion will remain in the soil, adsorbed by the clay fraction or enclosed into soil aggregates (Merbach and Wittenmayer 2004; Rasse et al. 2005), or incorporated into microbial biomass (Bottner et al. 1999; Kaštovská and Šantrůčková 2007).

Since the biochemical structure of harvest residues (C_{RE}) will not differ significantly from adjacent aboveground plant material, humification coefficients for this pool (H_{RE}) are similar to straw or green manure coefficients, depending on the specific crop. The humification coefficient for green manure has preliminarily been defined as $H_{GM} = 0.126$ based on Hülsbergen (2003). With straw, the humus reproduction coefficient is calculated based on the N content of the substrate. As the C:N ratio of straw is much wider than that of SOM, we assume that N is the limiting factor with humus reproduction and calculate H_{Straw} on the basis of the CN ratio of the substrate, which is in line with results of Nicolardot et al. (2001). Preliminarily, we define:

$$H_{\text{Straw}} = \frac{CN_{\text{SOM}}}{\text{SPC} : \text{SPN}} \quad (7)$$

H_{Straw} = humification coefficient for straw (factor).
 CN_{SOM} = C:N ratio of soil organic matter (ratio),
 default value $CN_{\text{SOM}} = 10.5$.
 SPC = side product C (kg C).
 SPN = side product N (kg N).

Sensitivity analysis

Sensitivity of the model output to parameter changes were assessed in twelve test runs based on an artificial crop rotation. With each test run, one single parameter was changed by 10% of the initial value, respectively. Sensitivity was analysed with descriptive methods only, a statistical measure was not applied.

Model validation

The model was tested for its ability to assess the relation between different cropping systems at a defined site with regard to their specific impact on SOM levels. To do so, humus balances were calculated plotwise for different treatments in two long-term field experiments, and the variation of humus balances within each field experiment was compared to the variation of SOC and STN topsoil level changes in the same survey period, respectively.

Data

Validation of the model was conducted in two long-term field experiments: the “Organic Arable Farming Experiment Gladbacherhof” (OAFEG) and field “C” of the DOK experiment (DOK).

The OAFEG (Schmidt et al. 2006) is located at Villmar (Hesse, Germany) in the North-Western Taunus hills. The soil is a haplic luvisol, mean annual precipitation reaches about 670 mm, and mean annual temperature is about 9.3°C.

In the two-factorial experiment, three organic farming systems are displayed with different crop rotation and fertilization. Further, four treatments with different tillage intensity are included. However: as tillage is not considered in humus balances including the HU-MOD algorithm up to now, we only included the plots of the reference tillage system (ploughing,

30 cm), resulting in a total plot number of $n = 12$ for model validation (3 treatments $\times$ 4 replications). A short description of the farming system treatments in the OAFEG is given in Table 6. The survey period in the OAFEG was 1998–2008.

The DOK experiment (Mäder et al. 2002) is located at Therwil, Switzerland. Soil is again a haplic luvisol, but both mean annual precipitation (785 mm) and temperature (9.5°C) are somewhat higher than at the OAFEG site.

In the DOK experiment, three treatments are established displaying different organic and conventional farming systems at two levels of fertilization intensity, and two additional treatments with zero resp. pure mineral fertilization. The crop rotation does not vary between treatments.

For reasons of capacity, only five out of eight treatments were included into the model validation, and only one out of three fields per treatment, respectively. Selection of the treatments and field was done without knowledge of data. Each treatment/field was replicated four times, and all replications were included into the evaluations ($n = 20$).

A description of the included treatments can be found in Table 6. The survey period in the DOK was 1978–2006.

Model validation was conducted at two levels of calibration in the OAFEG: initial calibration (INI) with consideration of crop yield levels (main and side product) and fertilizer C and N input, and improved calibration (IMP) with additional consideration of actual N contents in aboveground plant material with all crops including cover crops, differentiated according to main and side product (straw).

With the DOK experiment, only INI was possible for reasons of data availability in this study.

Annual data on soil organic carbon (SOC) and soil total nitrogen (STN, interpreted as soil organic nitrogen SON) contents in topsoils, taken plot by plot, allowed for a comparably good assessment of SOM level development in both LTFE. SOM level development (SOM_{trend}) was calculated based on linear trends of SOC and STN contents, respectively. SOM_{trend} was expressed in $\text{kg SOC ha}^{-1} \text{a}^{-1}$. Thus, SOC trend data could be included directly, while STN trend values were converted from $\text{kg STN ha}^{-1} \text{a}^{-1}$ to $\text{kg SOC ha}^{-1} \text{a}^{-1}$ applying a default ratio of C:N = 11.6.

SOM levels were converted to SOM masses applying a default bulk density of 1.5 g cm^{-3} soil.

Table 6 Description of long-term field experiment treatments included into the model validation

Factor	Treatment ID	Treatment description					
		Fertilization	Crop rotation (cropland ratio)				
			Quotient				
			FL	GL	CE	RC	CC ^a
<i>Organic arable farming experiment Gladbacherhof (OAFEG)</i>							
Farming system	Mixed farming (MF)	Farmyard manure, no straw- and green manuring.	0.33	0	0.50	0.17	0.33 ^a
	Stockless farming with rotational ley (SFL)	Straw- and green manuring (all non-cash crops and all side products).	0.17	0.17	0.50	0.17	0.50 ^a
	Stockless farming without rotational ley/ intense organic cash crop farming (SFC)	Straw- and green manuring (all non-cash crops and all side products).	0	0.33	0.50	0.17	0.66 ^a
<i>DOK experiment (DOK)</i>							
Farming system	NOFERT	Without fertilization.	0.29	0.14	0.29	0.29	0.43 ^a
	CONMIN	Mineral fertilizer.					
	CONFYM	Mineral fertilizer and farmyard manure (slurry and solid stable manure)					
	ORGFYM	Farmyard manure (slurry and solid stable manure)					
	BIODYN	Composted farmyard manure, biodynamic preparations					

FL perennial fodder legumes, GL grain legumes, CE cereals, RC row crops, CC cover crops

^a Values for cover crops relate to the proportion of years with cover cropping in complete crop rotations

For reasons of data availability there has not been a correction according to different bulk density up to now.

Validation method

The ability of HU-MOD to assess the impact of management changes on SOM levels was evaluated by correlating humus balances and SOM level changes in different scenarios under consistent basic environmental conditions. To do so, we calculated HBA and linear changes of SOC and SON for several plots and treatments in two long-term field experiments. It was postulated that basic environmental conditions within each field experiment were consistent, and that treatments can be used as different management scenarios. As HRC in HU-MOD are dependent on yield levels, an additional evaluation on the plot level was conducted.

The statistical method applied was regression analysis, with HBA being the independent and SOM level change being the dependent variable.

Basic measures for method validation were the coefficient of determination (r^2), and the standard error

of the estimate (SEE). Due to the neglected impact of initial SOM levels, neither the intercept of the regression, nor the regression coefficient (b) can be considered as validation parameters, but they are reported for illustration of the observed correlations.

We conducted separate calculations applying SOC trend and SON trend as SOM level change indicators, respectively. Further, the different calibration levels in the OAFEG (see above) allowed for the assessment of model quality at different levels of data availability.

Results

Sensitivity analysis

Figure 2 shows the effect of single parameter changes by 10% on HBA based on an artificial crop rotation. Basically, changes in input-related parameters had a considerably stronger impact than changes in the plant-related default parameters.

With the plant-related parameters, changes in N_{PB} and N_{Fix} imposed the strongest impact on humus

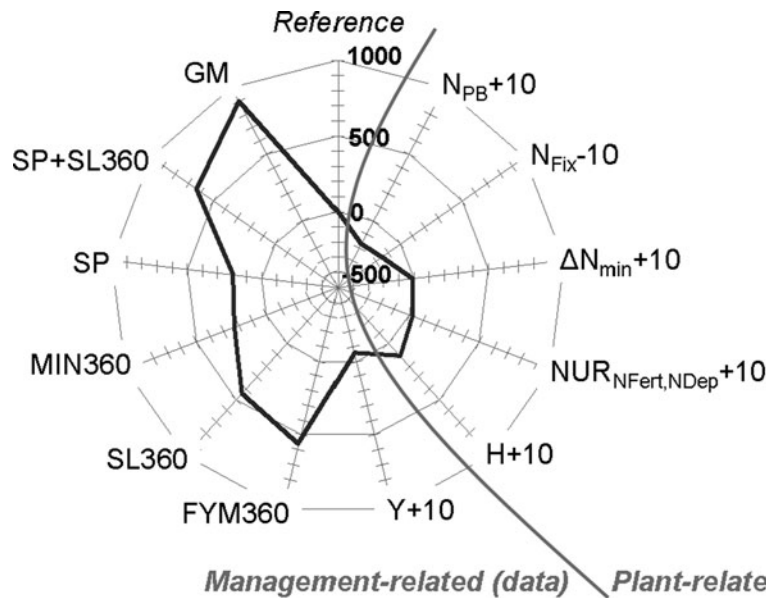


Fig. 2 Change of the humus balance for an artificial test crop rotation due to modification of single parameters. Changes in kg SOC ha⁻¹a⁻¹ relating to the crop rotation (6 fields: Clover Grass, yield (Y) 60 t FM ha⁻¹—Clover Grass, y 60 t FM ha⁻¹—Winter wheat, y 6 t FM ha⁻¹—Potatos, y 30 t FM ha⁻¹—Peas, y 3 t FM ha⁻¹—Winter rye, y 6 t FM ha⁻¹. No fertilization, all harvestable compartments removed. HBA = -564.5 kg SOC ha⁻¹a⁻¹. HBA in figure set to 0 as reference value!). N_{PB} + 10 = nitrogen in plant biomass increased by 10%, N_{Fix} - 10 = nitrogen derived from symbiotic fixation decreased

by 10%, ΔN_{min} + 10 = exceptional SON change increased by 10%, NUR_{NFert,NDep} + 10 = utilization rate for N from fertilizers with a high share of inorganic N, and from atmospheric deposition, H + 10 = humification rates increased by 10%, Y + 10 = yield increased by 10%, FYM/SL/MIN360 = solid farmyard manure/slurry/mineral fertilization corresponding to 360 kg N ha⁻¹ 6a⁻¹, SP = straw left on field, SP + SL360 = straw left + slurry fertilization corresponding to 360 kg N ha⁻¹ 6a⁻¹, GM = clover grass left as green manure (mulch)

balances (HBA, decrease by -170.4 kg SOC ha⁻¹a⁻¹ and -147.8 kg SOC ha⁻¹a⁻¹, respectively). Further, a 10% increase of humification coefficients increased HBA by +107.9 kg SOC ha⁻¹a⁻¹. The impact of changes in ΔN_{min} and NUR in the magnitude tested here were negligible with regard to crop rotation HBA.

Concerning the input-related parameters, fertilization showed a considerable impact on HBA (+201.3 kg SOC ha⁻¹a⁻¹...+885.3 kg SOC ha⁻¹a⁻¹), while a yield level increase of 10% had a comparably small effect on HBA (-63.2 kg SOC ha⁻¹a⁻¹). Out of the fertilizers, solid farmyard manure application had a considerably stronger positive impact on the crop rotation HBA (+555.0 kg SOC ha⁻¹a⁻¹) than slurry, mineral fertilization or side product (straw) left on the field (+201.3 kg SOC ha⁻¹a⁻¹...+438.5 kg SOC ha⁻¹a⁻¹). Straw + slurry fertilization had an even stronger impact (+639.8 kg SOC ha⁻¹a⁻¹) than solid farmyard manure application. However, the strongest impact on HBA again resulted from green manure

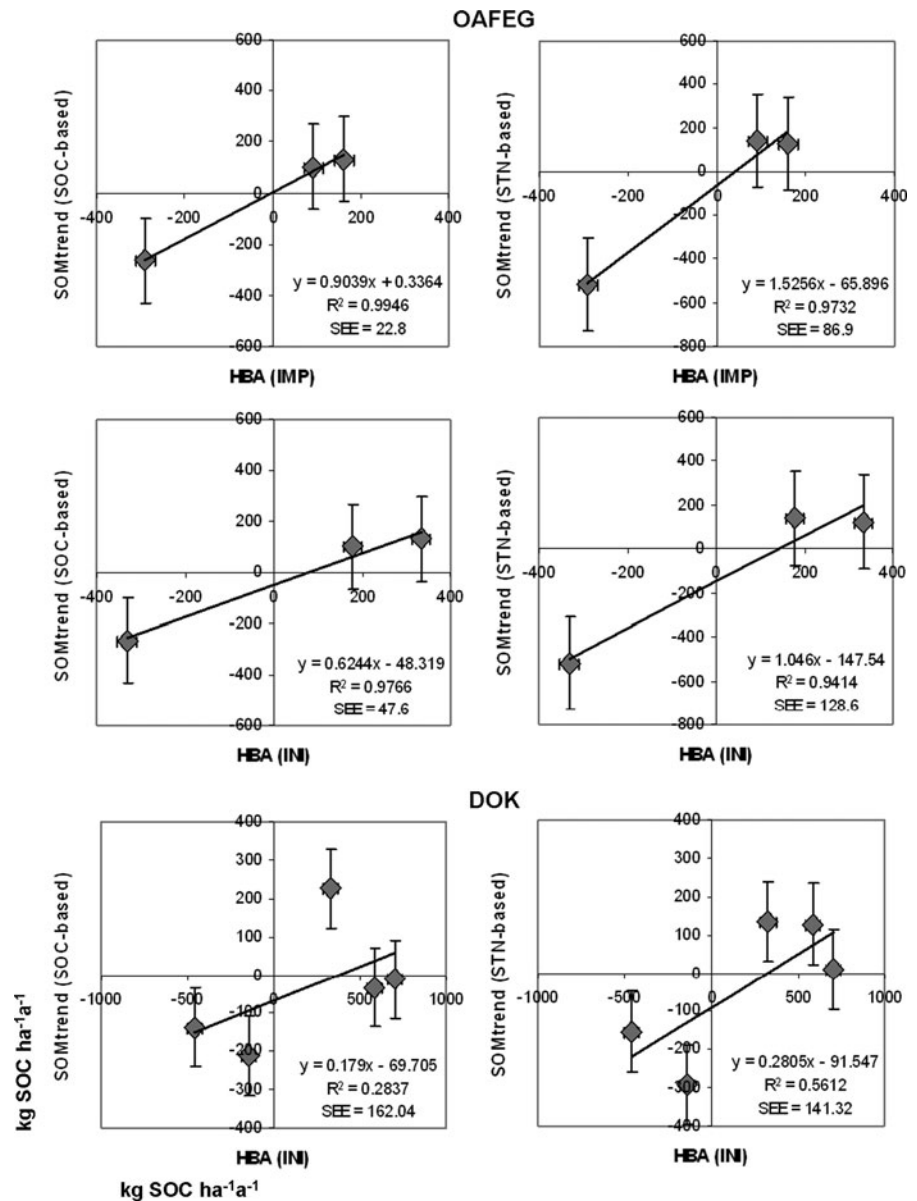
utilization of the clover grass, which pushed the crop rotation HBA up by +885.3 kg SOC ha⁻¹a⁻¹.

Validation

Results on the correlation between humus balances and SOM development in the two included LTFE on the treatment level are shown in Fig. 3. Basically, a positive correlation was observed with all validation runs. Coefficients of determination were on a comparable and high level (>0.90) with all validation runs in the OAFEG, and were considerably lower in the DOK, especially with the SOC-based validation.

Standard errors of the estimate on the other hand differed considerably between SOC and SON based validation in the OAFEG, as well as between initial and improved calibration. The total range of SEE in the OAFEG was 22.8...128.6 kg SOC ha⁻¹a⁻¹. SEE were lower with improved calibration.

Fig. 3 Correlation between humus balances and soil organic matter (SOM) level change variation in the OAFEG and DOK long-term field experiments on the treatment level. *Error bars* show the mean standard error of estimates. Evaluations referring to treatment level data, respectively, with $n = 3$ in the OAFEG and $n = 5$ in the DOK. SOM level change (SOM_{trend}) was calculated based on linear trends of soil organic carbon (SOC), and soil total nitrogen (STN), respectively. Survey period was 1998–2008 (OAFEG), and 1978–2006 (DOK). *INI* initial calibration, consideration of main and side product yield, and C and N input with fertilizers, *IMP* improved calibration, C and N contents of all harvestable plant material considered in addition to *INI*



In the DOK, SEE were 162.04 (SOC based) and 141.32 (SON based) and thus on a comparable level with both validation runs.

Data on model validation on the plot level is given in Table 7. In the table, intercept and b are reported for better comparability with Fig. 2. Due to a considerable spreading of single plot values, coefficients of determination are somewhat lower with the plot level compared to the treatment level in the OAFEG. Naturally, SEE were considerably higher on the plot than on the treatment level as well, exhibiting values

between 358.7 and 440.9 $\text{kg SOC ha}^{-1}\text{a}^{-1}$. In the DOK, neither r^2 nor SEE differed significantly between the plot and the treatment level.

Discussion

The basic principle of the HU-MOD algorithm is to relate crop yields to both SOM demand and SOM supply.

Table 7 Correlation between humus balance and soil organic matter (SOM) level change variation on the PLOT level in the OAFEG and DOK long-term field experiments

LTFE	SOM level change	Calibration level	Regression parameters			
			Intercept kg SOC ha ⁻¹ a ⁻¹	b	r ² proportion	SEE kg SOC ha ⁻¹ a ⁻¹
OAFEG	SOC-based	INI	-49.36	0.64	0.24	360.6
	STN-based	INI	-150.23	1.09	0.38	440.9
	SOC-based	IMP	0.59	0.92	0.25	358.7
	STN-based	IMP	-65.02	1.60	0.40	432.0
DOK	SOC-based	INI	-71.12	0.19	0.26	142.9
(Field C)	STN-based	INI	-146.86	0.27	0.41	143.0

Sample size was $n = 12$ (OAFEG), and $n = 20$ (DOK). Data refers to regression analyses with humus balance (independent) and SOM level change (dependent). SOM level change was calculated based on linear trends of soil organic carbon (SOC), and soil total nitrogen (STN), respectively. Survey period was 1998–2008 (OAFEG), and 1978–2006 (DOK)

Humus reproduction coefficients calculated according to HU-MOD show a negative correlation between crop yields of non-legumes and humus reproduction, if fertilization is held constant and below or equal to the N demand in the lowest yield level plot included into the comparison. The reason is that the increasing demand for N from SOM mineralization with increasing yield level cannot be compensated for by a higher residue supply. Even though the HU-MOD sub-algorithm for SOM supply calculation relates to C, nitrogen must be considered as a limiting factor for logical considerations, as N taken up by plants cannot be completely returned to soils by 100%. Of course, the assumption is only true if referring to a fixed soil C:N ratio. However, a negative correlation between non-legume crop yields and SOM development (with other factors held constant) could be shown in another paper (Brock et al. 2011), and is supported by results of Stevens et al. (2005), who found that crop uptake of soil N was above 50 percent of total N uptake even with optimal mineral N fertilization. Still, the HU-MOD approach does not contradict results on a positive effect of higher crop yield levels on SOM stocks, as reported by Grant et al. (2002), Halvorson et al. (1999), or Rasmussen and Parton (1994). Such results have usually been obtained comparing treatments with different N fertilization, and the observed positive effect in most cases has to be interpreted as a relative accumulation comparing the included treatments. As N supply by fertilization is accounted for in the HU-MOD SOM loss sub-algorithm, higher yield levels connected to higher N supply will lead to

relatively higher HRC, well in line with the results reported here.

While an overestimation of “humus demand” is counteracted by the consideration of N supply from different sources, it has to be examined whether a priming effect of excessive available N supply has to be accounted for in the algorithm, as indicated by results of Mulvaney et al. (2009), or Kuzyakov et al. (2000).

Validation of the model-based humus balance in two long-term field experiments showed a good performance of the model, especially on the treatment level.

Despite of the significantly longer survey period in the DOK, the model performed less good than in the OAFEG. This situation was mainly caused by two factors: first, the zero-fertilization treatment received a higher HBA compared to the mineral fertilizer treatment due to considerably lower crop yields, but exhibited a greater decrease of SOM levels (Fig. 2). Apparently, nitrogen utilization from the mineral fertilizer was much higher than assumed in the model. As stated in the model description, the parametrization of the nitrogen utilization rate for easily available fertilizer nitrogen (NUR_{NFert}) is based mainly on experimental results reported by Leithold (1982), Rauhe et al. (1987), and Hülsbergen (2003). However, utilization rates of nitrogen appear to be dependent on several factors and thus are highly variable (Küstermann et al. 2010; Walther et al. 2001). However, Bosshard et al. (2009) found an even lower utilization of mineral fertilizer N in the DOK mineral fertilizer treatment than assumed in HU-MOD.

The second factor of the modest performance of HU-MOD in the DOK is the comparably high increase in SOM levels in the biodynamic treatment despite of a lower HBA compared to the farmyard manure and farmyard manure + mineral fertilizer treatment. This situation is likely to point at either a higher true humification rate of the applied compost, and/or at a higher utilization of nitrogen from the compost.

Validation results do not indicate a major shortcoming of HU-MOD with regard to the rating of straw fertilization. However, the comparably high HBA for the SFL system not consistent with the negative values of the SOM development indicators may express the need for an adaptation of N_{Fix} and/or humification coefficients for green manure.

In HU-MOD, straw fertilization or green manure (non-legume green manure crops) will not produce positive HBA, if no N is supplied by symbiotic fixation and/or fertilization. Again the assumption is that SOM mineralization derived nitrogen in plant material cannot be fully returned to soils, implying an at least slightly negative Humus reproduction.

As most studies relate to C instead of N, a positive effect of straw fertilization has been reported as a result of the large organic matter input (e.g. Lemke et al. 2010; Thomsen and Christensen 2004; Asmus and Völker 1984). However, the positive effect of straw fertilization again has to be interpreted as a foremost relative accumulation compared to a treatment with straw removal, as becomes very obvious in Saffih-Hdadi and Mary (2008). Concerning the quantification of SOM supply with straw, Zimmer et al. (2005) suggest that SOM build-up by straw is dependent on the availability of N from additional sources.

As for straw, the quantification of the impact of green manure on SOM remains problematic due to the heterogeneity of results. While Dabney et al. (2001) in a review article identified evidence for considerable build-up of soil organic matter of cover crops, Shepherd (1999), Biederbeck et al. (1997) as well as Gerzabek et al. (1997) could not find such an effect. In a modelling approach, Blombäck et al. (2003) calculated a slight increase of SOM through catch crop incorporation inspite of the large biomass input.

A contribution of catch crops to SOM build-up can be assumed as a result of symbiotic N fixation (legumes) and/or transformation of soil mineral N that might otherwise have been leached (Urbatzka

et al. 2009). Both mechanisms are considered as sources for SOM build-up in the HU-MOD algorithm by lowering HD (Eq. 3). However, it has been reported that green manure material is rapidly decomposed even at low temperature (Breland 1994) and that green manuring, and mulching of legume-grass stands in particular, may even produce a considerable priming effect (Fontaine et al. 2003; Kuzyakov et al. 2000; Torstensson and Aronsson 2000; Wu et al. 1993), thus not contributing significantly to SOM build-up despite the large primary organic matter input.

In HU-MOD, crop specific N_{Fix} values are applied according to KTBL (2005) and Hülsbergen (2003). However, it has been reported that the proportion of N_{Fix} in legume biomass is highly variable dependent on several environmental and management factors (Ledgard and Steele 1992; Schipanski et al. 2010; Liu et al. 2010). The only factor presently recognized with N_{Fix} calculation in HU-MOD is nitrogen fertilization. For an optimal calibration of the HU-MOD algorithm it may of course be helpful to apply more differentiated N_{Fix} values instead of fixed standard values. However, these data will usually not be available when applying the model in farming practice. Further, it has to be clarified whether the assumption of a proportional decrease of N_{Fix} with an increase of readily available N can be applied as a general principle. While the conclusions of Möller et al. (2008) are supported by Schipanski et al. (2010), Carlsson and Huss-Danell (2003) did not find a clear effect of fertilization on N fixation in a meta-analysis approach. Another possible shortcoming of N_{Fix} accounting in the HU-MOD algorithm may be the neglected transfer of N_{Fix} from legumes to non-legumes in mixed stands (Høgh-Jensen 2006; Brophy et al. 1987), and further the neglected nitrogen recycling in mulched stands (Hatch et al. 2007).

Concerning model calibration, it became obvious that the improved calibration with recognition of measured nitrogen contents of all above-ground plant compartments improved model results compared to initial calibration. However: as SEE were rather low on the treatment level of the OAFEG with both calibration levels, the model is likely to perform well under practice conditions with a typically rather limited data base. On the other hand it must be recognized, that even with the initial calibration, true main product yields were recognized with any crop. It will be tested in further model evaluation whether HU-

MOD will perform sufficiently with application of default values only.

Up to the present, tillage is not considered a factor in the HU-MOD algorithm. The reason was that the impact of tillage on SOM is still controversially discussed, and clear evidence for a positive impact of reduced tillage is still lacking (Baker et al. 2007). However, a bad performance of the model analysing SOM development with regard to different tillage systems in the OAFEG (data not shown) assumably must be interpreted as a hint towards the necessity of tillage impact recognition in humus balancing, as supported by results of Balesdent et al. (2000).

Conclusions

The humus balance model HU-MOD proved to be an applicable simple tool for the comparison of management systems in arable farming with regard to the impact on SOM levels. Even though an absolute quantification of SOM level changes is not possible due to the methodical approach bypassing the need for any data on soil parameters, the model may be used to assess a positive or negative impact of a management system or management period compared to a reference and thus may be used to assess the impact of management changes, or to analyse a specific impact for different management periods on a defined spatial unit.

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